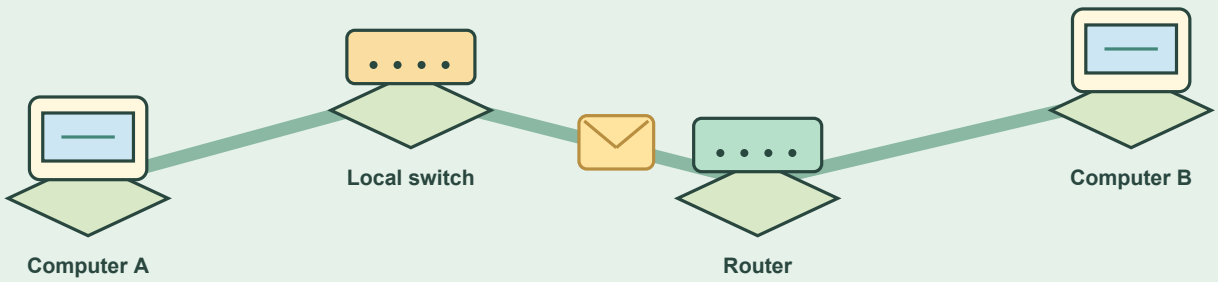
A FRIENDLY FIELD GUIDE

Your complete guide to PacketVerse.

Every section, every useful control, and a simple way to learn - even if computer networks are new to you.



Your words become small parcels with delivery labels.

THREE ROOMS	WHAT YOU DO THERE
CN Classroom	Learn Computer Networks Units 1-5 through pictures and short stories.
Network Lab	Send a message, follow packets, inspect devices, and try failures.
OSI Explorer	Explore seven layer worlds, the complete journey, and the shared data-format guide.

START WITHOUT MEMORISING ANYTHING

Open the lab, choose **The first hello**, type a message, leave **TCP** selected, and click **Send message**. The full walkthrough starts on page 3.

Open PacketVerse
The live links work on the computer where the local website is running. This PDF can be read offline.

YOUR ROAD MAP

Find what you need.

Read in order for a complete tour, or click a topic below to jump straight to it.

OPEN A TOPIC	PAGE
Your first message, the map and playback	3
The notebook and event timeline	4
Ready-made experiments and fault controls	5
The six guided lab missions	6
CN Classroom: studying Units 1-5	7
OSI Explorer: the seven layer worlds	8
All 7 together: the complete journey	9
The colourful data-format reference	10
Settings, sounds and saving experiments	11
Troubleshooting, shortcuts and a study routine	12

Choose your route

1. **For your syllabus:** start in CN Classroom, choose a unit, and watch one concept story.
2. **For hands-on learning:** open Network Lab and try a guided mission.
3. **For the seven-layer picture:** use OSI Explorer, then All 7 together.

TWO KINDS OF LEARNING

The Classroom and OSI Explorer use teaching illustrations. The Network Lab calculates packet journeys in a teaching model. Coverage labels tell you which classroom ideas the lab implements.

A few words you will meet

Packet: a small parcel of network data. **Hop:** one move to the next device. **ACK:** a delivery receipt. **Payload:** contents inside a wrapper. **Header:** a label before those contents.

NETWORK LAB / GET STARTED

Send your first hello.

Open the Network Lab

1. Choose **The first hello** in **Your experiment**.

2. Keep **Computer A** as the sender and **Computer B** as the receiver.

3. Type **Hello, network!** in the message box.

4. Keep **TCP** selected and click **Send message**. Playback starts.

5. Watch the moving parcels. Wait for your message and the completion status below the map.

Read the little world

OBJECT	EVERYDAY MEANING
Computer	Someone sending or receiving the message.
Switch	A local sorting desk that directs frames to a connection.
Router	A junction that helps packets reach another network.
Cable / link	A road between neighbouring devices.
Moving parcel / ACK	Network data or a control message; an ACK acts like a receipt.

Take control of playback

CONTROL	USE IT TO...
Play / Pause	Run or freeze the animation.
Next event	Advance one recorded action; this is not always a whole hop.
Replay	Watch the same experiment, including its existing faults.
Timeline slider / speed	Move to another moment or change how quickly you watch.
Reset / Send message	Clear the current journey / run the current configuration.

Map key explains the colours: blue TCP data, purple UDP data and mint ACKs. Playback speed changes viewing speed, not the calculated result.

NETWORK LAB / LOOK INSIDE

Let the notebook translate.

Click a device, a moving parcel, or an event. The notebook explains the selected object or moment.

NOTEBOOK PAGE	THE QUESTION IT ANSWERS
Journey	What is happening right now?
Encapsulation	How is the message wrapped for delivery?
Packet Fields	Which addresses, numbers and labels are in this parcel?
Routing Tables	Which directions are known for reaching other networks?
MAC Tables	Which switch connection leads to which device?
ARP Cache	Which local IP addresses have known MAC labels?
TCP Timeline	How did the connection start? Which receipts or retries occurred?
Metrics	How much arrived, how much was lost, and how long did it take?
Why This Happened	What caused this particular action?

A simple inspection routine

1. Send a message and pause it partway through.

2. Click a parcel, device, or event in the log. Use the device buttons below the map if that is easier.

3. Choose a page from **Notebook page**. Expand **What do these words mean?** for plain-language definitions.

4. Click **Follow live** when you want the notebook to follow playback again.

WHY DOES THE NOTEBOOK LOOK STUCK?

A selected event is deliberately pinned. It can show an older snapshot while the animation continues. Click **Follow live** to return to the current moment.

Event timeline lists actions in time order. **Packet list** is another way to select packet-related events. You can expand the notebook with its expand icon, or show/hide it with the header Notebook button.

NETWORK LAB / EXPERIMENT

Change one thing. See why.

Choose an experiment, then click Send message. Ask: what changed, what did the network do, and did the whole message arrive?

EXPERIMENT	WHAT TO LOOK FOR
The first hello	Normal successful delivery.
A parcel goes missing	TCP retries missing data.
The missing receipt	Recovery after an acknowledgement is lost.
Taking the express lane	Missing UDP data without automatic recovery.
The scenic route	Delivery through an alternative router path.
An island in the network	Failure when the available routed paths are broken.
Out of order, in good hands	TCP handles pieces that arrive out of order.
A packet with a short life	The packet exhausts its router-hop allowance.
A slower stretch of cable	Extra link delay makes the journey take longer.

TCP or UDP?

TCP tracks delivery and byte order and can retry missing data. **UDP** sends datagrams without TCP's delivery tracking or automatic retry. Compare **A parcel goes missing** with **Taking the express lane**.

Introduce a fault yourself

1. Start a TCP journey and pause it before it finishes.

2. Click **Drop next data**, then resume. Watch for the loss, a wait, and a retry.

3. Open **...** for more faults: drop an ACK, disable/enable a link, or change a link delay and click **Apply**.

FAILURE CAN BE THE CORRECT LESSON

A failed message in a failure scenario does not mean the website is broken. Use **Why This Happened**. If playback has already finished, replay or send a new message before introducing a fault.

NETWORK LAB / GUIDED MISSIONS

Six small discoveries.

Click Guided lab missions in the lab. Each mission prepares an experiment and gives you one objective.

MISSION	WHAT YOU NEED TO DO
1 Message to Bytes	Complete delivery, then inspect **Encapsulation**.
2 A Local Introduction	Complete delivery, then inspect **ARP Cache** and **MAC Tables**.
3 Beyond the Neighborhood	Send the prepared experiment and watch delivery through **Router 3**.
4 A Reliable Conversation	Watch TCP lose data, retry, and complete delivery.
5 No Promises, Just Packets	Observe the prepared UDP loss without automatic retries.
6 The Road Back	Observe the failed journey and inspect **Routing Tables**.

How to finish a mission

1. Choose a mission and read its objective in the banner.

2. Click **Send message**. Keep the prepared scenario while completing the mission.

3. Let the journey reach its end. Open the notebook pages the mission asks you to inspect.

4. Look for **Discovery made** with a check mark. Your completed discoveries are remembered in this browser.

IF IT STILL SAYS IN PROGRESS

Check that playback reached the end and that you opened the required notebook pages. Some missions intentionally finish with failed delivery: explaining the failure is the objective.

Return to free experimentation

Use **Leave lesson** to leave the mission. You can then choose another scenario, type another message, or try faults yourself.

The missions belong to the Network Lab. They complement the five-unit classroom rather than replacing your syllabus.

CN CLASSROOM / YOUR SYLLABUS

Learn one picture at a time.

Open CN Classroom - Computer Networks theory Units 1-5.

UNIT / ON-SCREEN NAME	MAIN IDEA
1 The big picture	Networks, connections and the organisation of communication.
2 One local crossing	Local-link frames, errors and access to the shared medium.
3 Find the next hop	Addresses and directions between networks.
4 A reliable conversation	Communication between applications and traffic handling.
5 Services we use	Naming, web communication, email and network management.

Use a concept story

1. Choose your **unit**, then click a **concept**.

2. Click **Play story**. Follow the picture and read its short caption.

3. Use **Next step**, the previous-step arrow, or replay when you want more time.

4. Read the takeaway. Open extra concept notes only when you need them.

5. Expand **Worked example & revision**. Try explaining an answer aloud before revealing the self-check.

Understand the coverage labels

LABEL	WHAT IT MEANS
Live model	The relevant modelled behaviour can be explored in the lab.
Partial model	The lab demonstrates only part of the concept.
Theory only	A visual explanation is provided; the packet lab does not implement it.

Open network lab prepares a related example; click **Send message** after it opens. **Full notes** downloads study notes. Unit 1 also links to OSI Explorer. Close the classroom with the close button or Escape.

OSI EXPLORER / LAYER WORLDS

Seven jobs, one network.

Open Layer worlds - choose a numbered layer to explore its job.

LAYER	REMEMBER ITS JOB LIKE THIS
7 Application	Give applications useful network services.
6 Presentation	Represent information so both ends can interpret it.
5 Session	Help keep a conversation together.
4 Transport	Deliver to the correct app; TCP also tracks ordered delivery.
3 Network	Use network addresses to reach the destination.
2 Data Link	Deliver a frame across the local link.
1 Physical	Carry bits using physical signals.

Explore each illustrated world

1. Type a message in the message box.

2. Click **Play this layer** and follow the highlighted stations and parcel.

3. Click a station to inspect its job. **Next station** lets you move manually.

4. Open **A little more detail** when you want the explanation.

5. Use **Next world**, **Previous world** or the numbered layer buttons to move between pages.

SOFTWARE JOBS, TOO

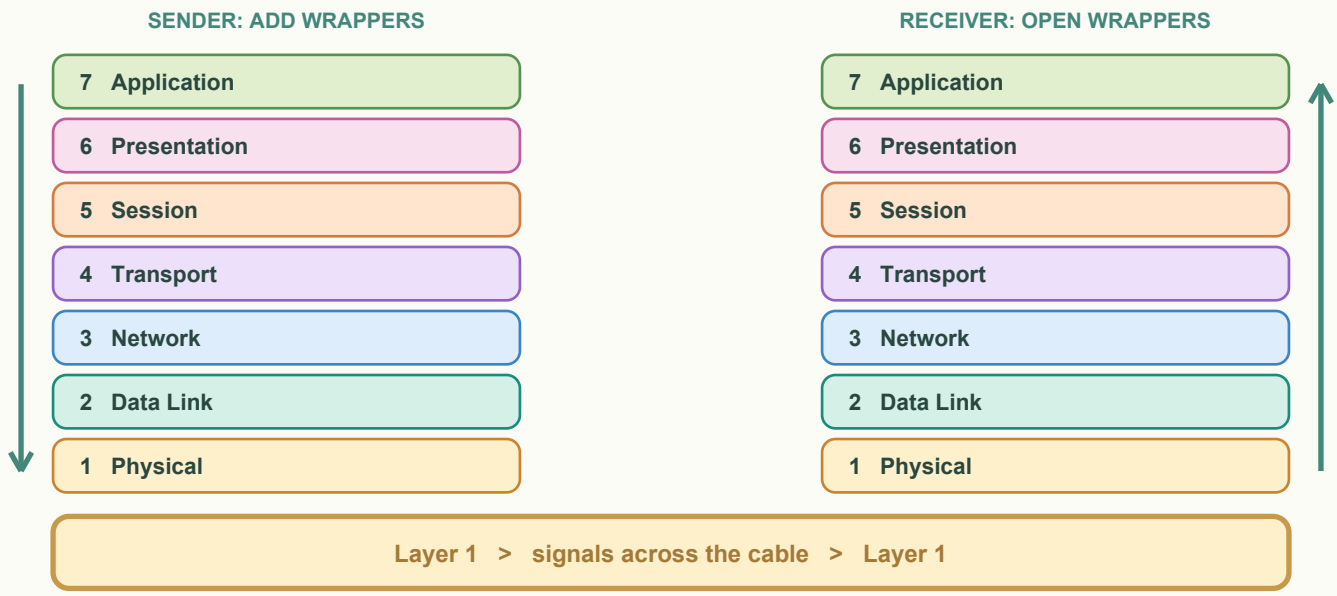
Layers 5-7 show software roles inside the sending and receiving computers. Their stations illustrate jobs, rather than separate physical boxes you need to connect.

Inside this explorer, **Layer worlds**, **All 7 together** and **Data formats** are related views. The OSI views share their example message. The Network Lab has its own message input.

OSI EXPLORER / ALL 7 TOGETHER

Follow the whole journey.

Open All 7 together - see the two computers' stacks side by side.



1. Enter your message and click **Play whole journey**.

2. Watch the highlight move down the sender's stack. TCP, IP and Ethernet wrappers are added.

3. Follow the signals across the link, then watch the receiver open the wrappers on the way up.

4. Look for your original message at the receiving application.

WHY ARE THERE 15 STEPS?

The same seven layers are used at both ends: 7 sender steps + 1 link step + 7 receiver steps. There are still only seven OSI layer roles.

Explore at your own pace

Click a layer on either computer, the cable, or a progress marker to jump to that stage. Use **Previous**, **Next handoff**, **Speed** and **Replay** to control the walkthrough.

Handoff means one job passing the bundle to the next. **Explore this layer's world** opens a closer look. Editing the message starts this illustrated journey again from the beginning.

OSI EXPLORER / SHARED REFERENCE

Same hello. Different formats.

Click Data formats in either OSI view. Read each row from left to right: layer, data-unit name, format picture and example.

LAYER	DATA UNIT	A SIMPLE FORMAT PICTURE
7 Application	Data	Request line + headers + message body
6 Presentation	Data	Text represented as UTF-8 bytes
5 Session	Data	Application bytes associated with conversation context
4 Transport	TCP segment	TCP header + application bytes; UDP uses a datagram
3 Network	IP packet	IP header + transport segment
2 Data Link	Frame	MAC header + IP packet + error-checking trailer
1 Physical	Bits	The frame's bits carried as signals

Three words that unlock the table

Header: a label before the contents. **Payload:** the contents inside the wrapper. **Trailer:** extra information at the end; Ethernet's example includes an error check.

Try your own message

1. Edit the message in the reference panel. The OSI example updates too.

2. Compare its text, UTF-8 byte preview and bit preview. For example, **cafe** uses 4 UTF-8 bytes, while **café** uses 5.

3. Expand the Transport row for TCP and UDP field layouts. Open the Network and Data Link rows for IP and Ethernet fields.

4. Close the panel with the close button or **Escape** to return to the same explorer view.

READ THE LABELS, NOT JUST THE COLOURS

The highlighted row matches the current layer. Some byte and bit previews are labelled excerpts. Layers 5-7 do not each add a universal extra header. On a phone, swipe inside the table to see its other columns.

SITE TOOLS / SETTINGS

Make the lab comfortable.

Click the settings gear. Keep the network values at their defaults until the normal journey makes sense.

SETTING	EVERYDAY MEANING
Random seed	A repeatability setting for random behaviour.
Loss percentage	How often transmitted IP frames may be lost.
Initial IP TTL	The packet's router-hop budget.
Teaching MSS	The teaching model's message-chunk size; smaller chunks create more pieces.
TCP timeout	How long TCP waits before a retry.
Retry limit	How many retries are allowed.

Use **Delay the first application segment** for an out-of-order example. After changing network settings, click **Save settings**, then send a message.

Sound and motion

The header's music-note button enables or mutes effects. **Effects volume** in Settings adjusts loudness. Audio starts after an interaction. Packet, layer, discovery and fault cues remain; typing and ordinary clicks stay quiet. Use **Reduce motion** for a less animated experience.

Keep an experiment

1. **Export scenario:** download the experiment setup as a JSON file.
2. **Import JSON:** load a saved setup, then click **Send message**.
3. **Clear saved data:** remove stored preferences and learning progress when you want a fresh start.

WHAT IS REMEMBERED?

Learning progress and preferences are saved in this browser. OSI views also share a saved example message. To keep a particular lab setup, export it; do not rely on a running animation being restored after a refresh.

QUICK HELP / KEEP GOING

When something feels confusing.

Try the smallest helpful action first. You can always return to The first hello for a normal experiment.

WHAT YOU NOTICE	WHAT TO DO
Notebook shows an older moment	Click **Follow live**.
Mission stays In progress	Let playback end and inspect its required notebook pages.
Delivery fails in a failure scenario	Read **Why This Happened**; the failure may be intentional.
Fault button cannot affect a finished run	Replay or send another message first.
No sounds	Check the music-note button, Settings volume and device volume.
Map is wider than the phone screen	Swipe the map or use the station buttons.
Table is wider than the phone screen	Swipe inside the table; layer labels stay at the left.
Too many unfamiliar words	Open the optional plain-language help and notebook definitions.

Keyboard shortcuts

KEY	ACTION
Tab	Move between controls.
Enter / Space	Activate a focused button.
Space outside a form or dialog	Toggle Network Lab playback.
Escape	Close a dialog.

A comfortable study routine

Choose one classroom concept > watch its picture > try a related lab experiment > inspect the notebook > explain the result in your own words.

YOU DO NOT NEED EVERY CONTROL AT ONCE

Learn one small journey at a time. Start with a successful hello, then try a missing parcel, then explore why the layers cooperate.